

CHAPTER 7: LIGHT

Part A: Chapter Notes

Section 7.1: Light and Its Propagation

- Light is a form of radiant energy essential for sight. Objects become visible when reflected or emitted light from them enters the eye [1].
- Light originates from luminous bodies (self-emitting, e.g., Sun, stars, fireflies) or is reflected by non-luminous bodies (e.g., Moon, planets) [1, 2].
- Thermal classification divides sources into hot sources (light emitted due to elevated temperature, e.g., incandescent filaments) and cold sources (light emitted without heat rise, e.g., neon lamps) [2].
- Geometric classification includes point sources (aperture size negligible, e.g., pinholes) and extended sources (larger dimensional light sources) [3].
- Mediums affect propagation: transparent (full transmission), translucent (partial transmission), and opaque (zero transmission) [3, 4].
- Light travels in straight lines—a phenomenon called rectilinear propagation [7].

Equations, Formulas & Units

- Speed of light in vacuum/air: $c \approx 3 \times 10^8 \text{ m s}^{-1}$ [6].

Key Examples

- Luminous organisms: Electric fish, select species of deep-sea fish, and insects (fireflies) [2].
- Cold light sources: Neon bulbs, sodium vapour lamps, fluorescent tubes [2].

Section 7.2: Shadows and Eclipses

- Shadows are formed due to the rectilinear propagation of light when an opaque obstacle blocks light rays [7, 9].
- A shadow is composed of two potential zones: the umbra (zone of complete illumination blockage) and the penumbra (zone of partial illumination) [9].
- A point source forms a sharp-edged shadow consisting purely of an umbra [9, 10].
- An extended source forms both an umbra and a surrounding penumbra [9, 11].

Effect of Distances and Source Dimensions [10, 12, 13]:

- Moving the light source closer to the object increases the size of both umbra and penumbra regions [12].
- Moving the screen away from the object expands both umbra and penumbra regions [12].
- If an extended source is larger than the opaque object, the umbral region tapers down to a cone and disappears entirely if the screen is placed far enough away [13].

Key Phenomena & Applications

- Tube lights cast faint, indistinct shadows because their large extended size produces vast penumbras and negligible umbras [13].
- High-flying aeroplanes cast no shadow on the ground because the umbra tapers off completely in mid-air before reaching the Earth's surface [13].

- Solar Eclipse: Occurs when the Moon passes directly between the Sun and Earth in a straight line, casting the Moon's shadow on Earth [13].
- Lunar Eclipse: Occurs when the Earth passes directly between the Sun and Moon, casting Earth's shadow onto the Moon [14].

Section 7.3: Reflection of Light and Plane Mirrors

Principle / Theory

- Reflection is the bouncing back of incident light rays into the same medium upon striking a surface [14].
- Regular reflection occurs on smooth, polished surfaces (e.g., plane mirrors), preserving parallel beam structures to form sharp visual images [15].
- Irregular (diffuse) reflection occurs on rough surfaces (e.g., cardboard), scattering rays in all directions so no image is formed [16].

Equations & Formulas

Laws of Reflection:

1. Incident ray, normal, and reflected ray lie in the same plane [19].
 2. Angle of incidence equals angle of reflection: $i = r$ [18, 19].
- Glancing angles: $g_i = 90^\circ - i$; $g_r = 90^\circ - r$. Since $i = r$, $g_i = g_r$.
 - Number of images (n) produced by two plane mirrors inclined at angle θ [27]: $n = \frac{360^\circ}{\theta} - 1$.
 - Periscope: Uses two plane mirrors fixed facing each other at 45° inside a Z-shaped tube (inner surface blackened) to shift light paths by 90° twice for viewing over obstacles [29, 30].
 - Kaleidoscope: Three equal plane mirrors arranged at 60° angles inside a tube containing loose colored glass bits form intricate, infinite symmetrical patterns upon rotation [30].

Part B: Six Key Questions and Answers

Question 1: Fundamental Concepts & Sources of Light

Q: Define optics. Differentiate between luminous and non-luminous bodies as well as hot and cold sources of light, giving examples of each from the text.

- Optics: Optics is the branch of physics which deals with the nature, sources, properties, and effects of light [1].
- Luminous vs. Non-Luminous Bodies:
 - Luminous bodies emit light on their own (e.g., the Sun, stars, electric fish, and certain species of insects like fireflies) [1, 2].
 - Non-luminous bodies do not emit light on their own but become visible when light falls on them and is reflected (e.g., the Moon, planets) [1, 2].
- Hot vs. Cold Sources of Light:
 - Hot sources emit light as a result of an elevation in temperature (e.g., incandescent filaments, candle flames) [2].
 - Cold sources emit light without a rise in temperature (e.g., fluorescent bulbs, neon bulbs, sodium vapour lamps) [2].

Question 2: Rectilinear Propagation & Shadow Formation

Q: What is the rectilinear propagation of light? Explain how the shadows formed by a point source differ from those formed by an extended source.

- Rectilinear Propagation: It is the property of light by which it always travels along a straight path in a homogeneous medium [3, 4]. This can be verified experimentally using three wooden uprights (P, Q, R) with aligned pinholes; if any one upright is moved slightly out of line, the light flame disappears from view [4].
- Shadow Formation:
 - Point Source: When an opaque body blocks light from a point source (such as light through a pinhole), it forms a dark patch called an umbra [5, 6].
 - Extended Source: When an extended light source (larger than a point source, e.g., a tube light or bulb) is used, both a dark central region (umbra) and a surrounding partially illuminated region (penumbra) are formed [5, 7, 8].

Question 3: Laws of Reflection & Glancing Angles

Q: State the two Laws of Reflection, and define the glancing angles of incidence and reflection.

- Laws of Reflection:
 - 1. The incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane [9].
 - 2. The angle of incidence is strictly equal to the angle of reflection ($\angle i = \angle r$) [9, 10].
- Glancing Angles:
 - Glancing Angle of Incidence (g_i): The angle subtended between the incident ray and the reflecting surface [10].
 - Glancing Angle of Reflection (g_r): The angle subtended between the reflecting surface and the reflected ray [10].

- Since $\angle i = \angle r$, the glancing angle of incidence is always equal to the glancing angle of reflection ($g_i = g_r$) [10].

Question 4: Characteristics of Images Formed by a Plane Mirror

Q: Distinguish between a real image and a virtual image, and list four primary characteristics of an image formed by a plane mirror.

- Real vs. Virtual Image:
 - A real image can be captured on a screen (e.g., images projected in a cinema theatre) and is formed by the actual intersection of reflected rays [11, 12].
 - A virtual image cannot be captured on a screen (e.g., image seen in a plane mirror) and is formed where divergent reflected rays appear to meet when produced backwards [11–13].
- 4 Characteristics of a Plane Mirror Image:
 - 1. It is of the same size as the object [14, 15].
 - 2. It is at the same distance behind the mirror as the object is in front of it ($v = u$) [12, 15].
 - 3. It is erect but laterally inverted (the left side of the object appears as the right side of the image) [12, 15].
 - 4. It is virtual [12, 15].

Question 5: Multiple Images in Inclined Mirrors (Numerical)

Q: State the formula to determine the number of images formed by two inclined plane mirrors. Calculate the number of images produced when two plane mirrors intersect at an angle of 60° .

- Formula: The number of images (n) formed by two plane mirrors inclined at an angle θ is given by: $n = 360^\circ/\theta - 1$ [16].
- Calculation:
 - Given angle of inclination $\theta = 60^\circ$ [17].
 - Substituting into the formula: $n = 360^\circ/60^\circ - 1 = 6 - 1 = 5$ images [16, 17].
 - Therefore, 5 images are formed [17].

Question 6: Optical Devices — Periscope & Kaleidoscope

Q: Describe the construction, working principle, and practical applications of a Periscope and a Kaleidoscope.

- Reflecting Periscope:
 - Principle & Construction: Works on the principle of the reflection of light [18]. It consists of a cylindrical tube bent twice at right angles containing two plane mirrors fixed facing each other at 45° angles to the tube walls [18]. The inner tube is blackened to prevent internal reflections [18].
 - Applications: Used in submarines to view surface ships and by soldiers in trenches to observe enemy movement over obstacles [19].
- Kaleidoscope:
 - Principle & Construction: Consists of three equal rectangular plane mirrors arranged inside a cylindrical tube at 60° angles to each other [19]. One end contains loose colored glass bits or paper pieces [19].
 - Working: Light undergoes multiple reflections between the inclined mirrors, creating beautiful, symmetrical, changing patterns upon rotating the tube [19].